

BPE Tokenization of the Same Sentence Across Languages (Case 2: Native Script Subwords)

(Fertility F = number of subword tokens produced per sentence)

English (Latin)

"The scientist discovered a new species"

The	scientist	discovered	a	new	species
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F = 6

German (Latin)

"Der Wissenschaftler entdeckte eine neue Art"

Der	Wissen	schaft	ler	ent	deck	te	eine	neue	Art
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F = 10

Amharic (Ge'ez)

"ሳይንቲስቱ አዲስ ዝርያ አገኘ"

(Tokenized in Native Ge'ez Script)

ሳይ	ንቲ	ስቱ	አ	ዲስ	ዝር	ያ	አ	ገኘ
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F = 9

Subword
segmentation

* Note: When the tokenizer includes Ge'ez vocabulary (e.g. Glot500/Qwen), tokens remain in native script. Without script merges (e.g. LLaMA), fertility surges to 38+ byte tokens.